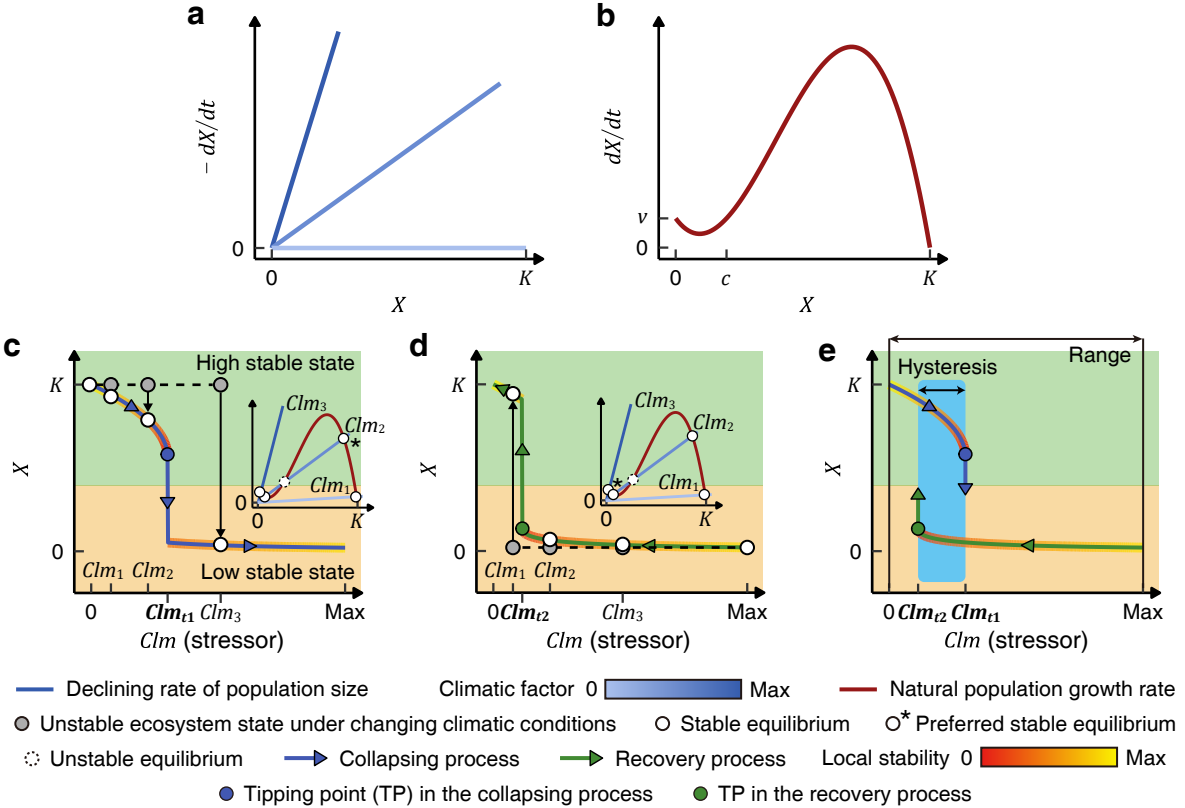
a
$-dX/dt$
0
0
K
X
b
dX/dt
v
0
0
c
K
X
c
High stable state
K
X
Clm_3
Clm_2
Clm_1
0
0
K
0
Clm_1 Clm_2
Low stable state
0
Clm_{t1} Clm_3
Max
Clm (stressor)
d
K
X
Clm_3
Clm_2
Clm_1
0
0
K
0
Clm_1 Clm_2
0 Clm_{t2}
Clm_3
Max
Clm (stressor)
e
Range
Hysteresis
K
X
0
0 Clm_{t2} Clm_{t1}
Max
Clm (stressor)
Declining rate of population size
Climatic factor 0
Max
Natural population growth rate
Unstable ecosystem state under changing climatic conditions
Stable equilibrium
Preferred stable equilibrium
Unstable equilibrium
Collapsing process
Recovery process
Local stability 0
Max
Tipping point (TP) in the collapsing process
TP in the recovery process